

RESTRAINTS

08
SECTION

OUTLINE 08-00
ON-BOARD DIAGNOSTIC.... 08-02

AIR BAG SYSTEM 08-10
SEAT BELT 08-11

08-00 OUTLINE

RESTRAINTS ABBREVIATIONS 08-00-1

RESTRAINTS FEATURES 08-00-1

RESTRAINTS ABBREVIATIONS

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DLC	Data Link Connector
DTC	Diagnostic Trouble Code
ELR	Emergency Locking Retractor
GND	Ground
PID	Parameter Identification
SAS	Sophisticated Air Bag Sensor

RESTRAINTS FEATURES

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Improved safety	• A driver-side air bag module has been adopted. • A passenger-side air bag module has been adopted. • A side air bag module has been adopted. • A pre-tensioner seat belt has been adopted.
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08-02 ON-BOARD DIAGNOSTIC

ON-BOARD DIAGNOSTIC FUNCTION
OUTLINE 08-02-1

ON-BOARD DIAGNOSTIC FUNCTION
[AIR BAG SYSTEM] 08-02-1

ON-BOARD DIAGNOSTIC FUNCTION OUTLINE

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- The air bag system has an on-board diagnostic function to facilitate the system diagnosis.
- The on-board diagnostic function consists of the following functions: a malfunction detection function, which detects overall malfunctions in the air bag system-related parts; a memory function, which stores detected DTCs; a display function, which indicates system malfunctions by DTC display; a PID/data monitoring function, which reads out specific input/output signals.
- Using the current diagnostic tool, DTCs can be read out and deleted, and the PID/data monitoring function can be activated.
- The system has a fail-safe function to prevent the accidental activation of the air bags in case of an air bag system malfunction.

ON-BOARD DIAGNOSTIC FUNCTION [AIR BAG SYSTEM]

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Self-Malfunction Diagnostic Function

- The self-diagnostic function determines that there is a malfunction, outputs a signal, as a DTC, to the DLC-2, and at the same time, flashes the air bag system warning light to advise the driver of a malfunction.
- The air bag system warning light illuminates or flashes to indicate a single DTC according to the present malfunction. (If there is more than one present malfunction, only one DTC will be displayed according to the preset priority ranking.)
- The air bag system warning light will flash the DTC pattern for five cycles, and then will remain illuminated until the engine switch is turned to the LOCK position.
- The self-diagnostic function consists of a present malfunction diagnostic and a past malfunction diagnostic.

DTC table

DTC					Malfunction location
Current diagnostic tool display	Air bag system warning light			Priority ranking	
	Flashing pattern				
B1231	13			3	SAS control module activation (deployment) control freeze
B1317	—	Air bag system warning light is illuminated all the time.		—	The SAS control module power supply voltage increases (16.1 V or more)
B1318	—	Air bag system warning light is illuminated all the time		—	The SAS control module power supply voltage decreases (less than 9 V)
B1342	12			2	SAS control module
	—	Air bag system warning light is illuminated all the time.		1	SAS control module (DTC 12 detection circuit malfunction)
B1869	—	Air bag system warning light is illuminated all the time.		1	Air bag system warning light system circuit open
	—	Air bag system warning light is illuminated does not illuminate.		—	Air bag system warning light system circuit short to ground
B1870	—	Air bag system warning light is illuminated all the time.		1	Air bag system warning light system circuit short to power supply
B1877	33			10	Driver-side pre-tensioner seat belt circuit resistance high
B1878					Driver-side pre-tensioner seat belt circuit short to power supply
B1879					Driver-side pre-tensioner seat belt circuit short to ground

ON-BOARD DIAGNOSTIC

DTC				Malfunction location
Current diagnostic tool display	Air bag system warning light		Priority ranking	
	Flashing pattern			
B1881	34		9	Passenger-side pre-tensioner seat belt circuit resistance high
B1882				Passenger-side pre-tensioner seat belt circuit short to power supply
B1883				Passenger-side pre-tensioner seat belt circuit short to ground
B1885	33		10	Driver-side pre-tensioner seat belt circuit resistance low
B1886	34		9	Passenger-side pre-tensioner seat belt circuit resistance low
B1913	19		8	Driver-side air bag module circuit short to body ground
	21		7	Passenger-side air bag module circuit short to body ground
B1916	19		8	Driver-side air bag module circuit short to power supply
B1925	21		7	Passenger-side air bag module system circuit short to power supply
B1932	19		8	Driver-side air bag module circuit resistance high
B1933	21		7	Passenger-side air bag module circuit resistance high
B1934	19		8	Driver-side air bag module circuit resistance low
B1935	21		7	Passenger-side air bag module circuit resistance low
B1992	22		12	Driver-side side air bag module circuit short to power supply
B1993				Driver-side side air bag module circuit short to ground
B1994				Driver-side side air bag module circuit resistance high
B1995				Driver-side side air bag module circuit resistance low
B1996	23		11	Passenger-side side air bag module circuit short to power supply
B1997				Passenger-side side air bag module circuit short to body ground
B1998				Passenger-side side air bag module circuit resistance high
B1999				Passenger-side side air bag module circuit resistance low

ON-BOARD DIAGNOSTIC

DTC			Malfunction location	
Current diagnostic tool display	Air bag system warning light			
	Flashing pattern	Priority ranking		
B2228	19		8	Driver-side air bag module assembly incorrect
B2230				
B2232				
B2234				
B2229	21		7	Passenger-side air bag module assembly incorrect
B2231				
B2233				
B2235				
B2444	43		6	Driver-side side air bag sensor system (internal circuit disabled)
B2445	44		5	Passenger-side side air bag sensor system (internal circuit disabled)
B2867	31		4	Poor connection of any SAS control module connectors
U2017	43		6	Driver-side side air bag sensor (communication error)
U2018	44		5	Passenger-side side air bag sensor (communication error)

PID/Data Monitoring Function

- By using the PID/data monitoring function, the monitored item of the input/output signal, as set on the SAS control module, can be freely selected and read out in real-time.
- An current diagnostic tool is used to read out PID/data monitor information.

PID name (definition)	Unit/condition	Condition/specification	terminal
CONT_RCM (Number of continuous DTC)	—	- DTC is detected: 1—255 - DTC is not detected: 0	—
CRSH_ST_D1 (Driver-side air bag sensor communication state)	OK/ FAULT	- Sensor normal: OK - Sensor communication error: FAULT	2Z, 2AA
CRSH_ST_D2 (Driver-side air bag sensor circuit state)	OK/ FAULT	- Sensor normal: OK - Sensor internal circuit error: FAULT	2Z, 2AA
CRSH_ST_P1 (Passenger-side air bag sensor communication state)	OK/ FAULT	- Sensor normal: OK - Sensor communication error: FAULT	2B, 2C
CRSH_ST_P2 (Passenger-side air bag sensor circuit state)	OK/ FAULT	- Sensor normal: OK - Sensor internal circuit error: FAULT	2B, 2C
D_PTENSFLT (Driver-side pre-tensioner seat belt circuit state)	SQ_LOWRES OPEN SHRT_B+ SHRT_GND Normal	- Air bag module circuit resistance low: SQ_LOWRES - Related wiring harness circuit open: OPEN - Related wiring harness short to power supply: SHRT_B+ - Related wiring harness short to ground: SHRT_GND - Related wiring harness normal: Normal	2P, 2S
DABAGR (Driver-side air bag module resistance)	ohm	Under any condition: 1.5—3.7 ohm	1S, 1V
DR_PTENS (Driver-side pre-tensioner seat belt resistance)	ohm	Under any condition: 1.5—3.1 ohm	2P, 2S

ON-BOARD DIAGNOSTIC

PID name (definition)	Unit/condition	Condition/specification	terminal
DS_AB (Driver-side side air bag module resistance)	ohm	Under any condition: 1.4—3.2 ohm	2M, 2O
DS_AB_ST (Driver-side side air bag module circuit state)	SHRT_B+ SHRT_GND OPEN SQ_LOWRES Normal	- Related wiring harness short to power supply: SHRT_B+ - Related wiring harness short to ground: SHRT_GND - Related wiring harness circuit open: OPEN - Air bag module circuit resistance low: SQ_LOWRES - Related wiring harness normal: Normal	2M, 2O
DS1_STAT (Driver-side air bag module circuit state)	SHRT_GND SHRT_B+ OPEN SQ_LOWRES Normal	- Related wiring harness short to ground: SHRT_GND - Related wiring harness short to power supply: SHRT_B+ - Related wiring harness circuit open: OPEN - Air bag module circuit resistance low: SQ_LOWRES - Related wiring harness normal: Normal	1S, 1V
DSB_P_ST (On demand driver-side pre-tensioner seat belt circuit state)	SQ_LOWRES OPEN SHRT_B+ SHRT_GND Normal	- Air bag module circuit resistance low: SQ_LOWRES - Related wiring harness circuit open: OPEN - Related wiring harness short to power supply: SHRT_B+ - Related wiring harness short to ground: SHRT_GND - Related wiring harness normal: Normal	2P, 2S
IG_V_2 (System IG1 voltage value)	V	engine switch to ON position: B+	1W
OD_CRST_D1 (On demand driver-side side air bag sensor communication state)	OK/ FAULT	- Sensor normal: OK - Sensor communication error: FAULT	2Z, 2AA
OD_CRST_D2 (On demand driver-side side air bag sensor circuit state)	OK/ FAULT	- Sensor normal: OK - Sensor internal circuit error: FAULT	2Z, 2AA
OD_CRST_P1 (On demand passenger-side side air bag sensor communication state)	OK/ FAULT	- Sensor normal: OK - Sensor communication error: FAULT	2B, 2C
OD_CRST_P2 (On demand passenger-side side air bag sensor circuit state)	OK/ FAULT	- Sensor normal: OK - Sensor internal circuit error: FAULT	2B, 2C
OD_DAB1_ST (On demand driver-side air bag module circuit state)	SHRT_GND SHRT_B+ OPEN SQ_LOWRES Normal	- Related wiring harness short to ground: SHRT_GND - Related wiring harness short to power supply: SHRT_B+ - Related wiring harness circuit open: OPEN - Air bag module circuit resistance low: SQ_LOWRES - Related wiring harness normal: Normal	1S, 1V
OD_DSAB_ST (On demand driver-side side air bag circuit state)	SHRT_B+ SHRT_GND OPEN SQ_LOWRES Normal	- Related wiring harness short to power supply: SHRT_B+ - Related wiring harness short to ground: SHRT_GND - Related wiring harness circuit open: OPEN - Air bag module circuit resistance low: SQ_LOWRES - Related wiring harness normal: Normal	2M, 2O
OD_PAB1_ST (On demand passenger-side air bag module circuit state)	SHRT_GND SHRT_B+ OPEN SQ_LOWRES Normal	- Related wiring harness short to ground: SHRT_GND - Related wiring harness short to power supply: SHRT_B+ - Related wiring harness circuit open: OPEN - Air bag module circuit resistance low: SQ_LOWRES - Related wiring harness normal: Normal	1M, 1P
OD_PSAB_ST (On demand passenger-side side air bag sensor circuit state)	SHRT_B+ SHRT_GND OPEN SQ_LOWRES Normal	- Related wiring harness short to power supply: SHRT_B+ - Related wiring harness short to ground: SHRT_GND - Related wiring harness circuit open: OPEN - Air bag module circuit resistance low: SQ_LOWRES - Related wiring harness normal: Normal	2I, 2L
P_PTENSFLT (Passenger-side pre-tensioner seat belt circuit state)	SQ_LOWRES OPEN SHRT_B+ SHRT_GND Normal	- Air bag module circuit resistance low: SQ_LOWRES - Related wiring harness circuit open: OPEN - Related wiring harness short to power supply: SHRT_B+ - Related wiring harness short to ground: SHRT_GND - Related wiring harness normal: Normal	2G, 2J
PABAGR (Passenger-side air bag module resistance)	ohm	Under any condition: 1.4—2.9 ohm	1M, 1P
PS_AB (Passenger-side side air bag module resistance)	ohm	Under any condition: 1.4—3.2 ohm	2I, 2L

ON-BOARD DIAGNOSTIC

PID name (definition)	Unit/condition	Condition/specification	terminal
PS_AB_ST (Passenger-side side air bag sensor circuit state)	SHRT_B+ SHRT_GND OPEN SQ_LOWRES Normal	- Related wiring harness short to power supply: SHRT_B+ - Related wiring harness short to ground: SHRT_GND - Related wiring harness circuit open: OPEN - Air bag module circuit resistance low: SQ_LOWRES - Related wiring harness normal: Normal	2I, 2L
PS_PTENS (Passenger-side pre-tensioner seat belt resistance)	ohm	Under any condition: 1.5—3.1 ohm	2G, 2J
PS1_STAT (Passenger-side air bag module circuit state)	SHRT_GND SHRT_B+ OPEN SQ_LOWRES Normal	- Related wiring harness short to ground: SHRT_GND - Related wiring harness short to power supply: SHRT_B+ - Related wiring harness circuit open: OPEN - Air bag module circuit resistance low: SQ_LOWRES - Related wiring harness normal: Normal	1M, 1P
PSB_P_ST (On demand passenger-side pre-tensioner seat belt circuit state)	SQ_LOWRES OPEN SHRT_B+ SHRT_GND Normal	- Air bag module circuit resistance low: SQ_LOWRES - Related wiring harness circuit open: OPEN - Related wiring harness short to power supply: SHRT_B+ - Related wiring harness short to ground: SHRT_GND - Related wiring harness normal: Normal	2G, 2J

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08-10 AIR BAG SYSTEM

AIR BAG SYSTEM OUTLINE	08-10-1	DRIVER-SIDE AIR BAG MODULE	
AIR BAG SYSTEM STRUCTURAL		CONSTRUCTION/OPERATION	08-10-5
VIEW	08-10-2	PASSENGER-SIDE AIR BAG MODULE	
AIR BAG SYSTEM		FUNCTION	08-10-5
WIRING DIAGRAM	08-10-2	PASSENGER-SIDE AIR BAG MODULE	
SAS CONTROL MODULE		CONSTRUCTION/OPERATION	08-10-5
FUNCTION	08-10-3	SIDE AIR BAG MODULE FUNCTION ...	08-10-5
SAS CONTROL MODULE		SIDE AIR BAG MODULE	
CONSTRUCTION/OPERATION	08-10-3	CONSTRUCTION/OPERATION	08-10-6
DRIVER-SIDE AIR BAG MODULE		PRE-TENSIONER SEAT BELT	
FUNCTION	08-10-4	FUNCTION	08-10-6
		PRE-TENSIONER SEAT BELT	
		CONSTRUCTION/OPERATION	08-10-7

AIR BAG SYSTEM OUTLINE

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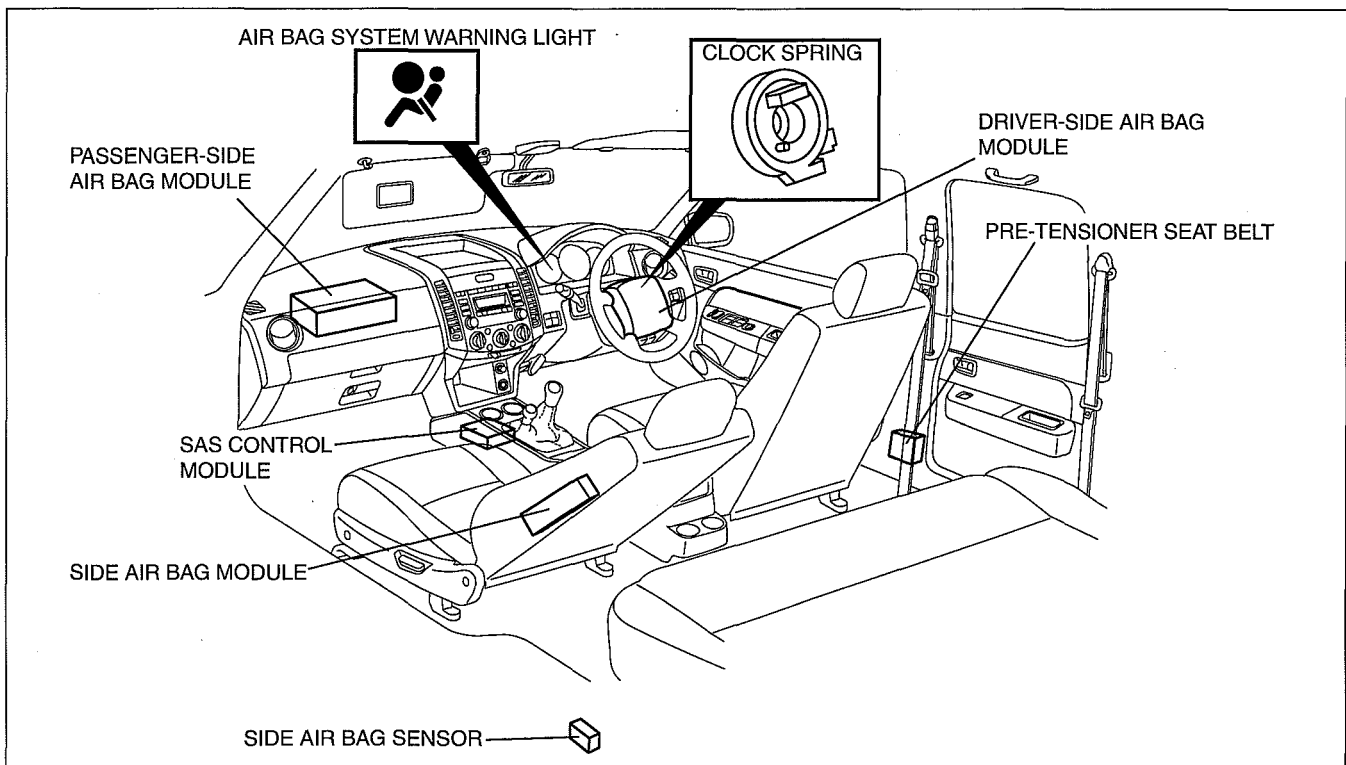
- The air bag system is a device that supplements the passenger restraint function of the seat belts. The air bag system will not have the designed effect if the seat belts are not worn properly.
- The air bag system is composed of the following parts:

Item	Outline
SAS control module	Recognizes actually equipped air bag module or pre-tensioner seat belt based on module configuration.
Side air bag sensor	Detects degree of impact, converts to an electrical signal, and sends the signal to the SAS control module. For operation, refer to SAS control module, Air Bag Module and Pre-tensioner Seat Belt Deployment Operation. (See 08-10-3 SAS CONTROL MODULE CONSTRUCTION/OPERATION)
Driver-side air bag module	Adopted to improve safety in frontal collisions.
Passenger-side air bag module	
Side air bag module	Chest-protection type side air bag module has been adopted.
Pre-tensioner seat belt	Piston-type pre-tensioner seat belt has been adopted.

AIR BAG SYSTEM

AIR BAG SYSTEM STRUCTURAL VIEW

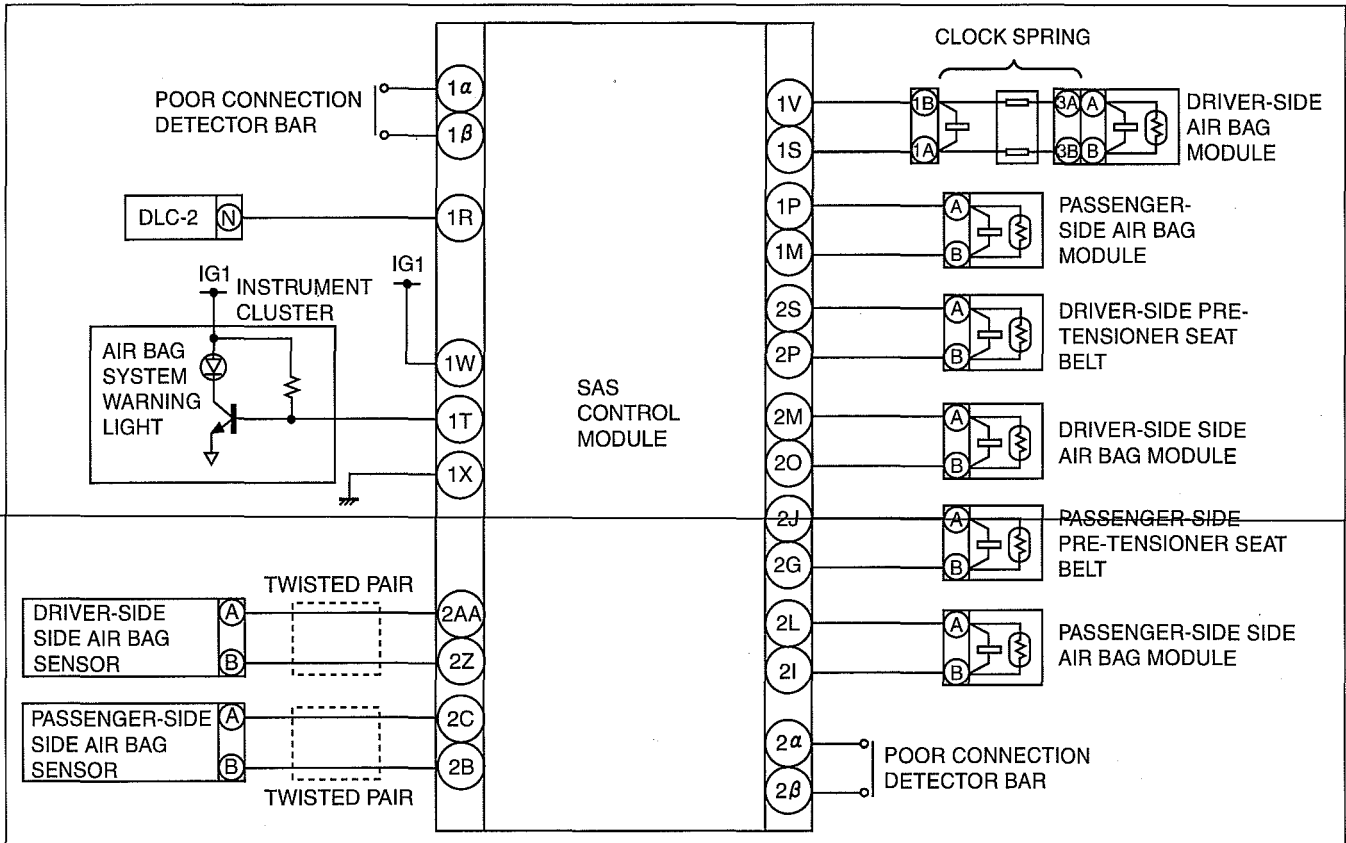
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AIR BAG SYSTEM WIRING DIAGRAM

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AIR BAG SYSTEM

SAS CONTROL MODULE FUNCTION

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Outline

Sensing function

- Safing and the crush sensors are built into the SAS control module.
- If the degree of impact detected by the safing and the crush sensors exceeds the set value during a collision, the SAS control module sends an operation (deployment) signal to the air bag module and the pre-tensioner seat belt.

Backup power supply function

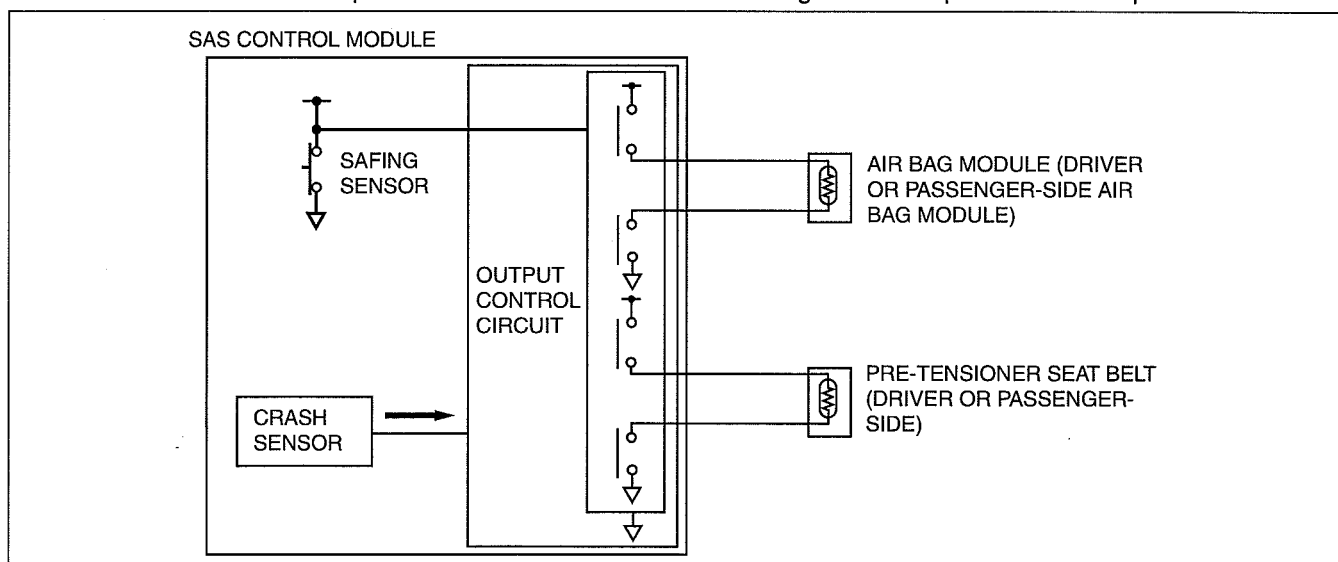
- The backup power supply function enables the condenser to discharge and supply power to assure air bag system operation/deployment properly for a specified time even if the power supply to the SAS control module is cut due to a collision.

SAS CONTROL MODULE CONSTRUCTION/OPERATION

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Air Bag System

1. The impact of a frontal or frontal offset collision to the vehicle is transmitted to the SAS control module.
2. The safing sensor inside the SAS control module detects the collision and activates. In addition, the crash sensor determines the impact and converts it to an electronic signal and outputs it to the output control circuit.

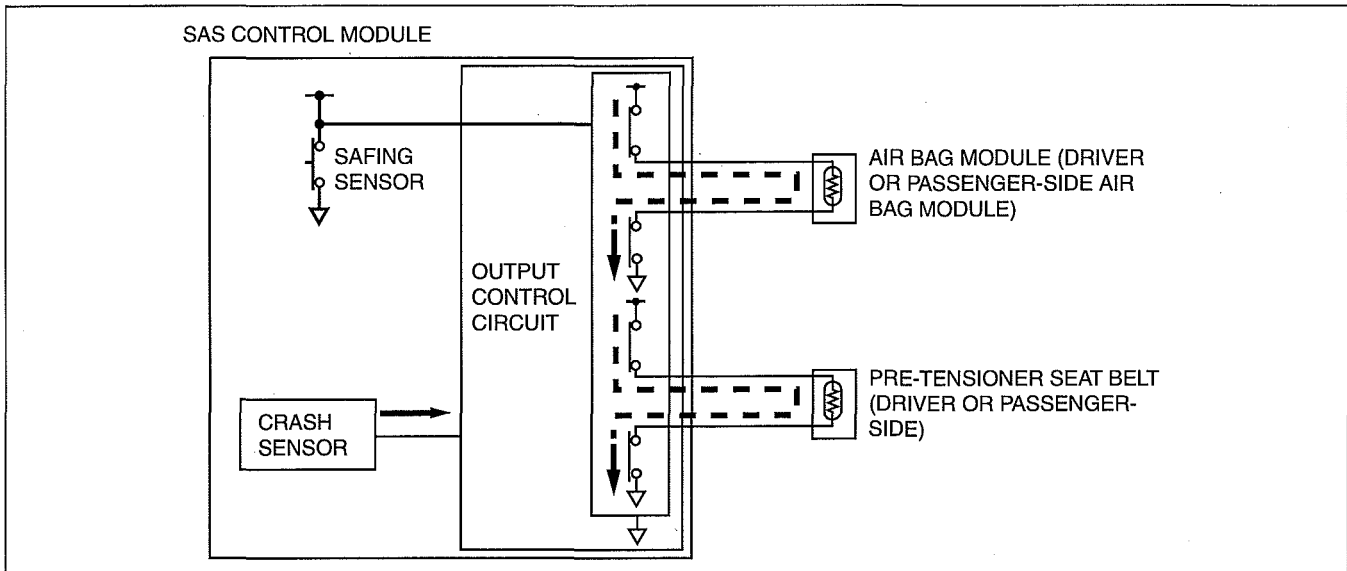


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3. The output control circuit calculates the electronic signals from the crash sensor and the valve is compared with the preset value. If the output signal received from the crash sensors exceeds the preset value, an ignition circuit is completed and an operation (deployment) signal is sent to both the driver and passenger-side air bag modules.

AIR BAG SYSTEM

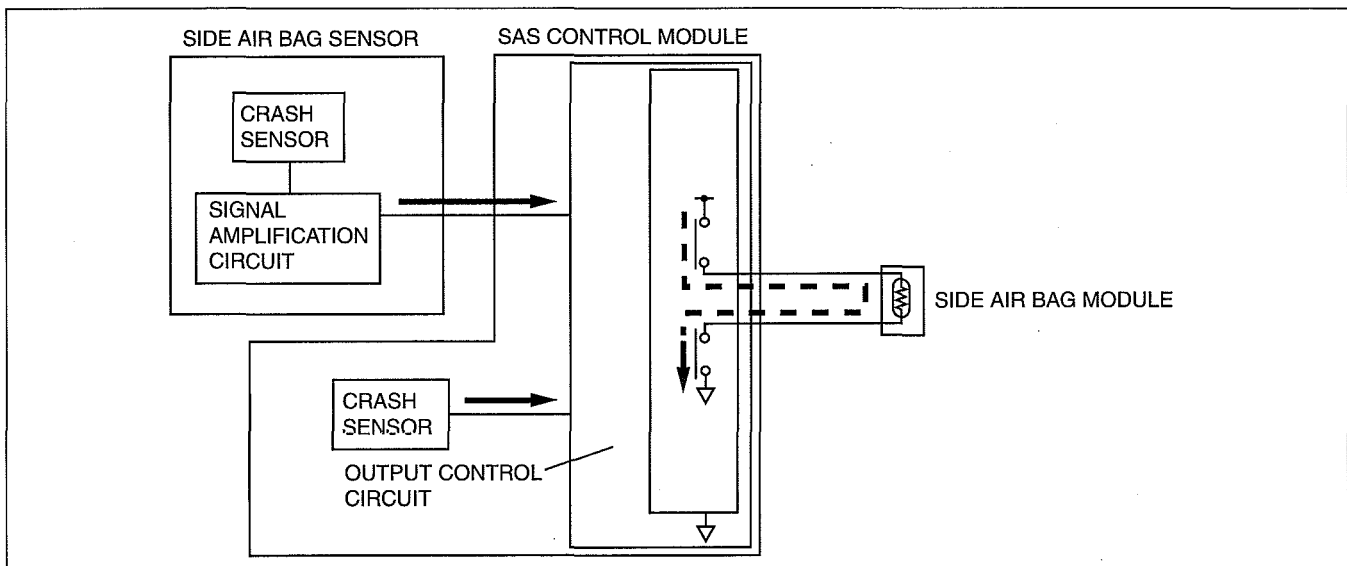
4. In conjunction with both the driver and passenger-side air bag module deployment, an ignition circuit for the pre-tensioner seat belt is completed and an operation (deployment) signal is sent to the pre-tensioner seat belt.



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Side air bag system

1. During a lateral collision to the vehicle, the crash sensors in the side air bag sensor and SAS control module detect the collision.
2. The degree of impact detected by the crash sensor in the side air bag sensor is converted to an electrical signal and sent to the SAS control module through the signal amplification circuit.
3. Simultaneously, the SAS control module crash sensor converts the degree of impact detected to an electrical signal.
4. The SAS control module processes the calculations for the two electrical signals at the output control circuit and compares the value to a preset value.
5. The output control circuit determines the degree of impact to the vehicle by the value from the crash sensors, completes a side air bag module ignition circuit, and sends the deployment signal to the air bag modules.



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DRIVER-SIDE AIR BAG MODULE FUNCTION

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Outline

- During a frontal or front offset collision, an operation signal from the SAS control module is received and the front air bag operates (deploys), softening the impact to the head and face areas of the driver.

AIR BAG SYSTEM

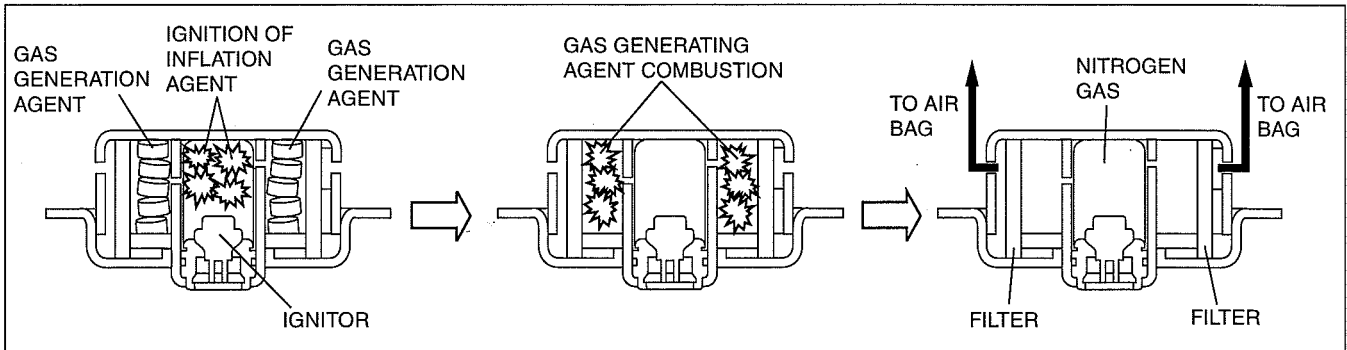
DRIVER-SIDE AIR BAG MODULE CONSTRUCTION/OPERATION

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- Installed in the center of the steering wheel.
- The inflator operates in the following order.

Inflator Operation

1. When an operation (deployment) signal is received from the SAS control module, the igniter built into the inflator generates heat and ignites the ignition agent.
2. The ignition of the ignition agent causes the combustion of a gas generating agent which forms nitrogen gas.
3. The nitrogen gas is cooled at the filter and the filtrate is injected into the air bag.



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PASSENGER-SIDE AIR BAG MODULE FUNCTION

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Outline

- During a frontal or front offset collision, an operation signal from the SAS control module is received and the air bag operates (deploys), softening the impact to the head and face areas of the front passenger.

PASSENGER-SIDE AIR BAG MODULE CONSTRUCTION/OPERATION

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- Installed in the dashboard.
- The inflator operation is the same as the driver's side air bag. (See 08-10-5 DRIVER-SIDE AIR BAG MODULE CONSTRUCTION/OPERATION.)

SIDE AIR BAG MODULE FUNCTION

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- During a collision to the side of the vehicle, the air bag operates (deploys) after receiving an operation signal from the SAS unit, defusing impact to the chest area of the driver and front passenger.

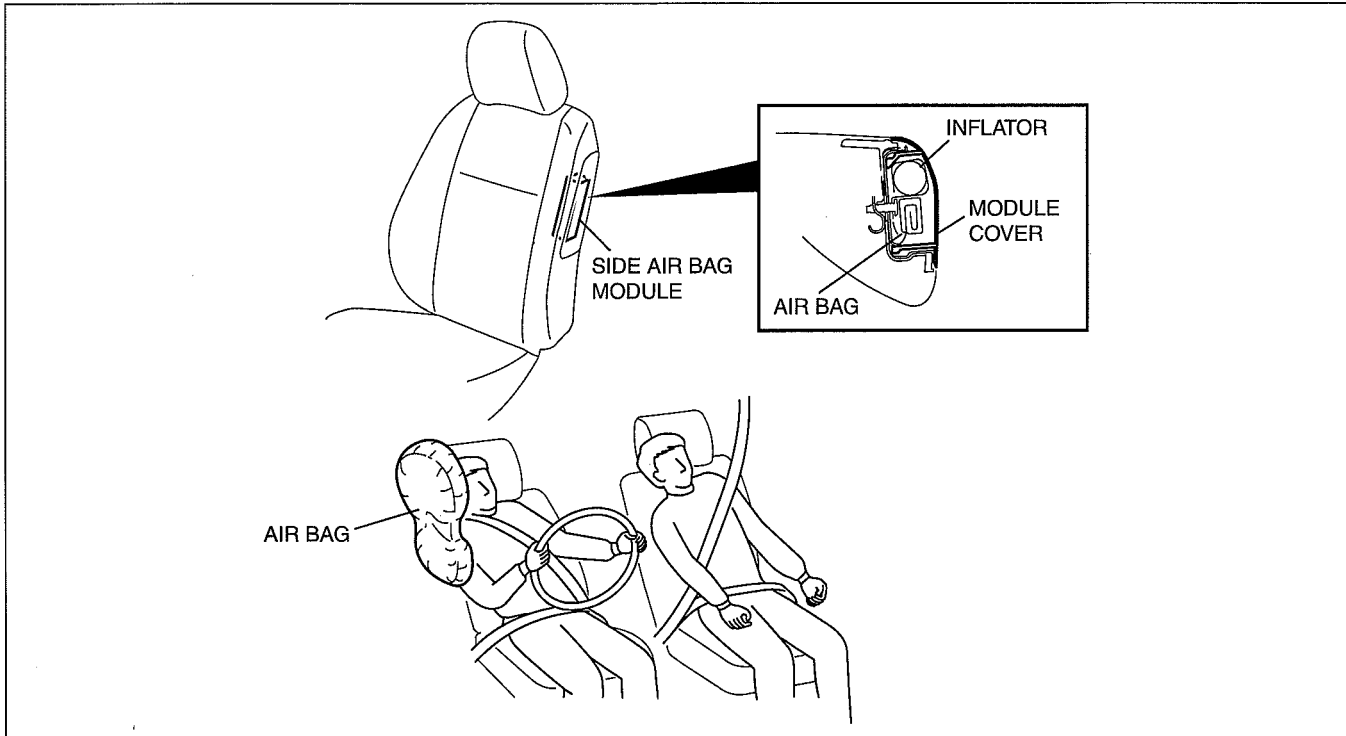
AIR BAG SYSTEM

SIDE AIR BAG MODULE CONSTRUCTION/OPERATION

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Construction

- Side air bag modules are installed on the outboard sides of the seat backs.
- The side air bag module is composed of an inflator, module cover and air bag.
- When an air bag deploys, the side air bag module cover is spread apart by the generation of argon gas from the inflator, inflating the air bag.



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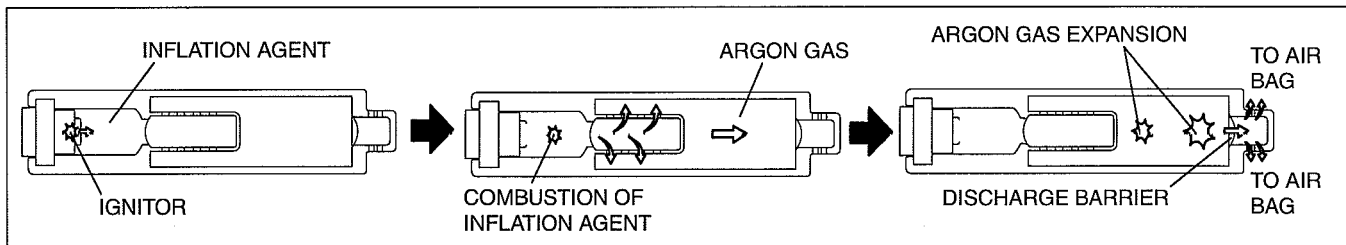
Operation

Air bag module deployment operation

- Refer to the SAS control module, Air Bag Module and Pre-tensioner Seat Belt Deployment Operation. (See 08-10-3 SAS CONTROL MODULE CONSTRUCTION/OPERATION.)

Inflator operation

1. The igniter built into the inflator begins to build up heat when the operation (deployment) signal is sent from the SAS control module. The inflation agent is ignited by the build up of heat in the igniter.
2. The argon gas expands due to the heat of the ignited inflation agent.
3. The expanding argon gas breaks the discharge barrier, is cooled and filtered by the filter, and then injected into the air bag.



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PRE-TENSIONER SEAT BELT FUNCTION

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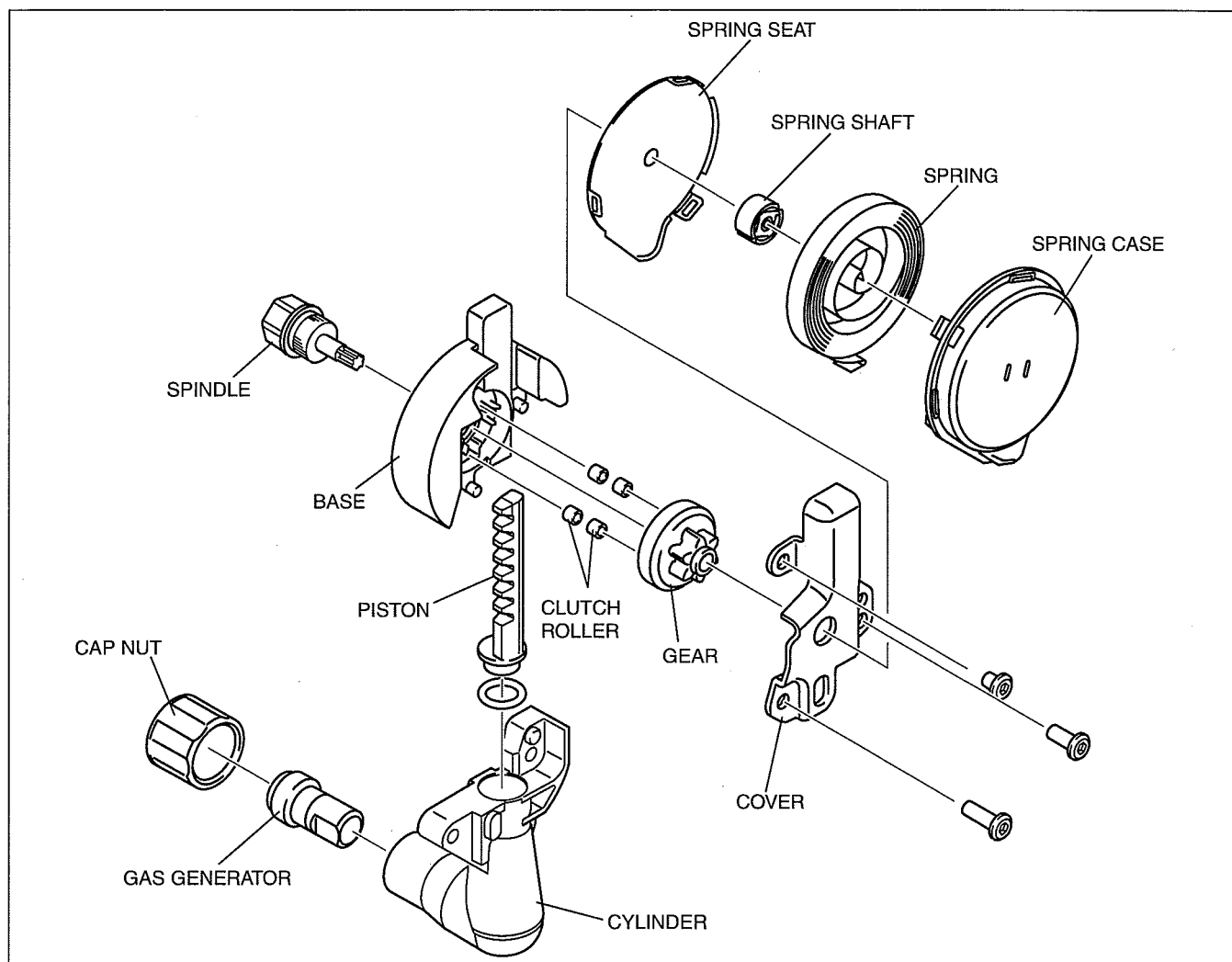
- When a vehicle is involved in a frontal or frontal offset collision and the seat belts are buckled, the pre-tensioner seat belt system receives an operation signal from the SAS control module, retracting and tightening the belt webbing instantly on the driver and passenger restraints.

AIR BAG SYSTEM

PRE-TENSIONER SEAT BELT CONSTRUCTION/OPERATION

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Construction



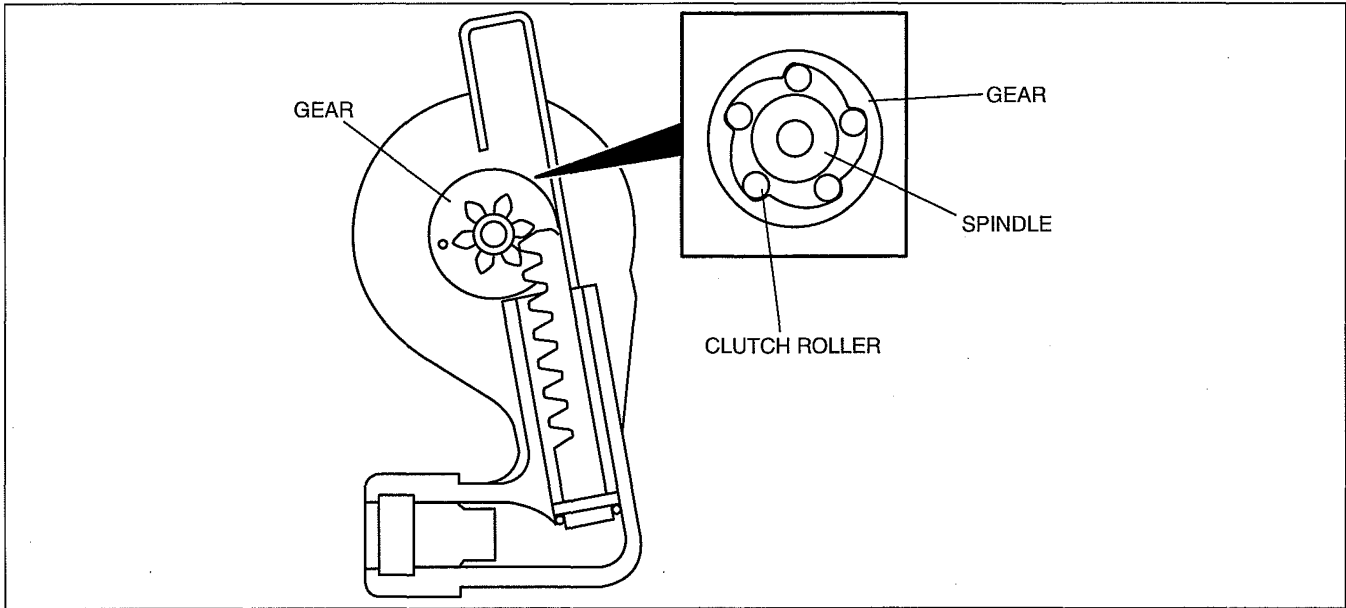
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AIR BAG SYSTEM

Operation

Normal (Seat Belt Pretensioners Not Operating)

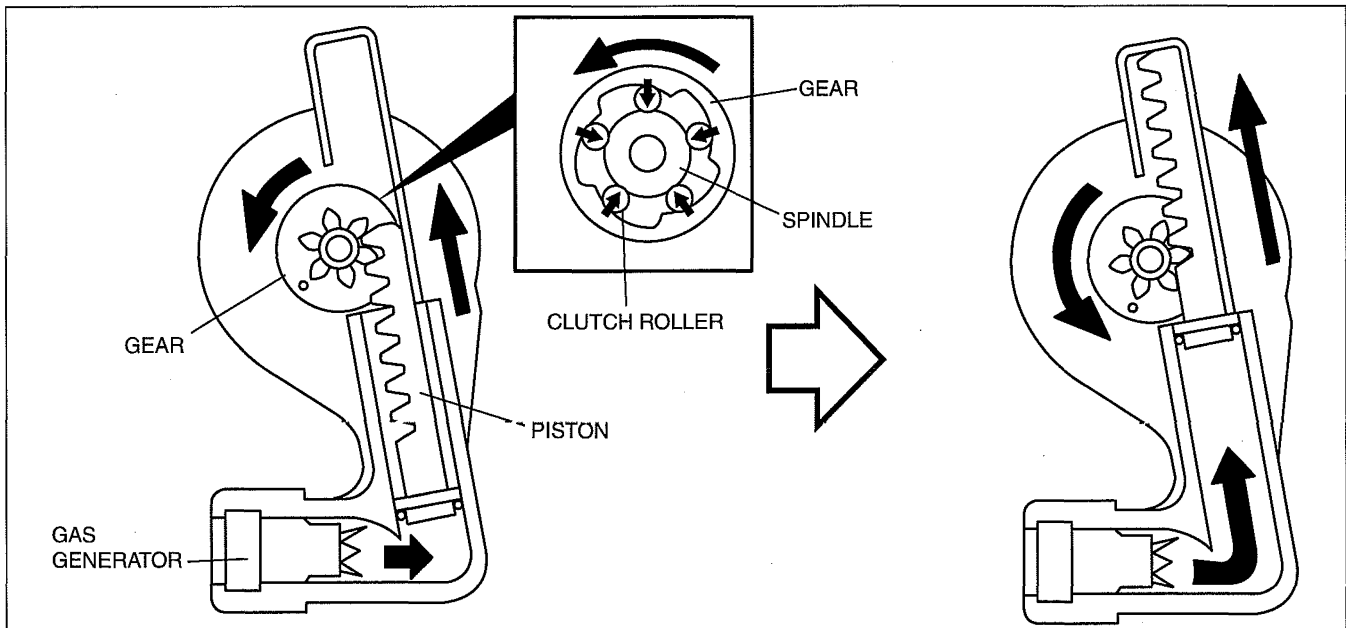
- Normally, the clutch roller installed to the outer circumference of the spindle sits in the recess of the gear and does not interfere with the spindle.
- The gear does not rotate when the belt is pulled or retracted because the spindle and gear are not engaged.



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Seat Belt Pretensioners Operating

1. When an operation signal is received from the SAS control module, the gas generator produces gas. Due to the pressure from the gas, the piston in the cylinder is pressed up.
2. The gear rotates while the piston moves up.
3. Based on the gear rotation, the clutch roller in the gear presses against the spindle. Due to this, the gear and spindle are engaged.
4. The belt is retracted in conjunction with the gear rotation.



EPU810ZT3013

08-11 SEAT BELT

SEAT BELT OUTLINE 08-11-1
SEAT BELT STRUCTURAL VIEW 08-11-1

LOAD LIMITER RETRACTOR
CONSTRUCTION/OPERATION 08-11-1

SEAT BELT OUTLINE

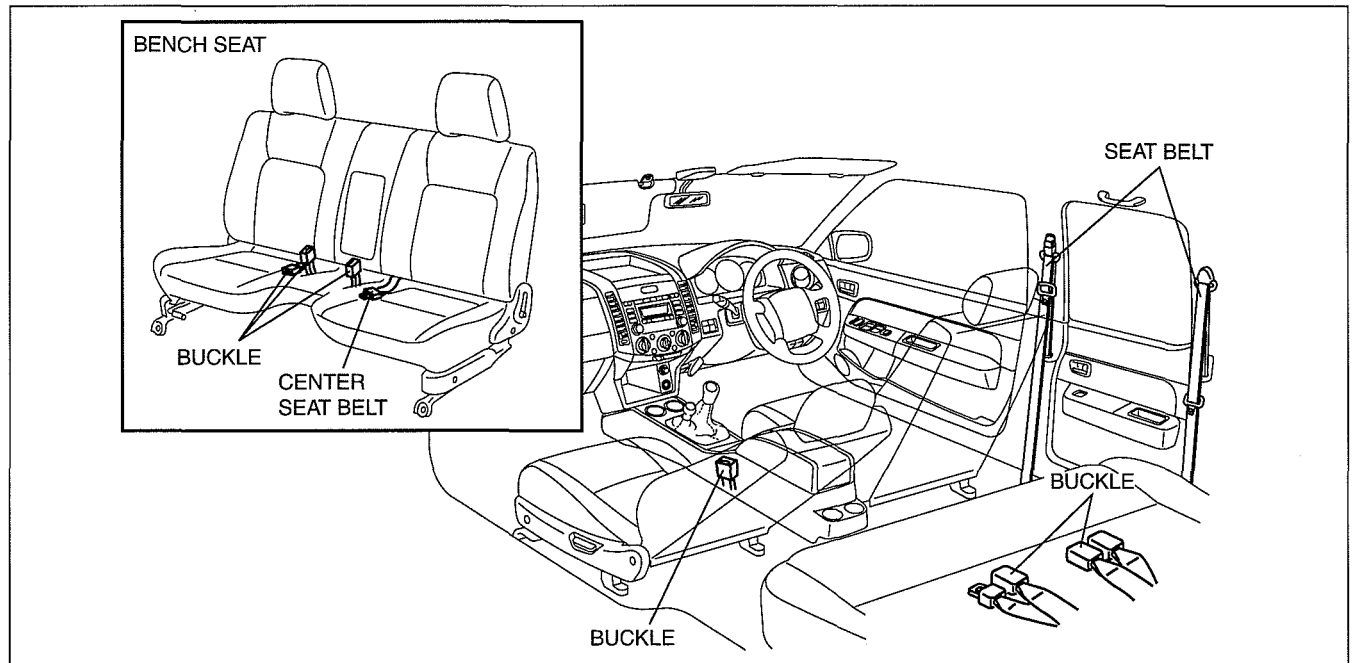
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Features

Improved safety	- Three-point seat belt with the following functions for driver and passenger seat adopted — ELR (Emergency Locking Retractor: emergency locking mechanism) — Pre-tensioner seat belt (vehicle with pre-tensioner seat belt) (See 08-10-7 PRE-TENSIONER SEAT BELT CONSTRUCTION/OPERATION.) — Load limiter, which adjusts restraint force of the seat belt to reduce the possibility of injury to passengers caused by excess seat belt pressure after pre-tensioner operation. (vehicle with pre-tensioner seat belt) - Three-point seat belt with the following functions for rear seat passengers adopted (Double cab) — ELR
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SEAT BELT STRUCTURAL VIEW

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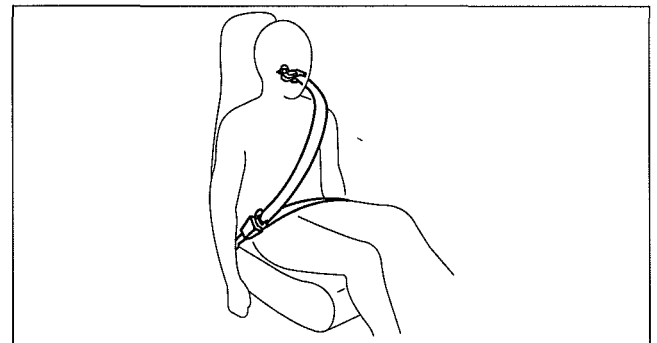


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LOAD LIMITER RETRACTOR CONSTRUCTION/OPERATION

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1. Initial state

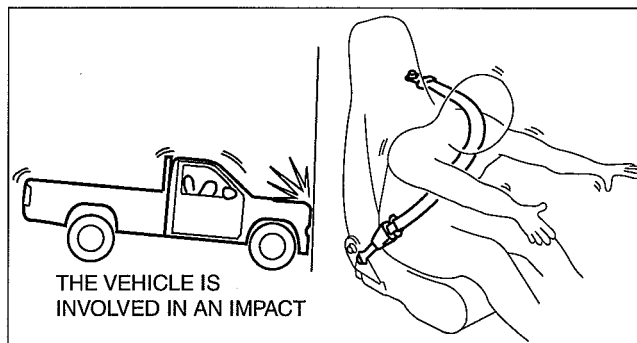


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SEAT BELT

2. ELR operation

When the force of impact is transferred to the belt, the retractor changes to the ELR condition, locking the belt and securing the passenger's body.



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3. Load limiter operation

After locking, if the force of impact transferred to the belt is strong enough to cause injury to the chest of the occupant, an adequate amount of belt is released to absorb the load applied to the chest.

